Dayananda Sagar College of Engineering

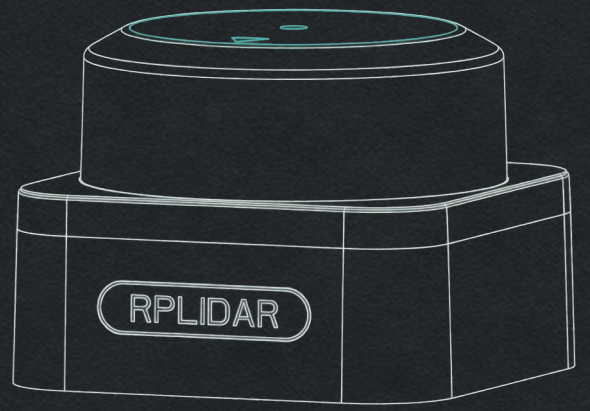
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Phase-I presentation / 22ETP76 / 2026-27

On-board SLAM for GPS-Denied Indoor Search



LiDAR reference illustration

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Introduction

Indoor search with limited sensing and compute

A useful payload must describe the space and surface acoustic cues that an operator can investigate.



SLAM

Estimate sensor motion while building a map of an unknown environment.



Acoustic sensing

Detect sounds of interest and estimate their bearing relative to the sensor.

Review scope: embedded SLAM, 3-D collaboration, loop closure and drone acoustics

Problem statement

Develop and evaluate a low-cost system that maps GPS-denied indoor spaces and provides acoustic cues for survivor search.

Combined sensing

Low-cost mapping needs a practical acoustic-search path.

Measured performance

Map appearance alone cannot establish accuracy.

Limited resources

Compute, power and rotor noise constrain the design.

Objectives

Targets for the complete project



01

Onboard indoor mapping

Develop a mapping pipeline suited to constrained sensing and compute.



02

Spatial and collaborative mapping

Extend planar scans to height maps and study offline 3-D / multi-agent methods.



03

Acoustic event detection and bearing

Characterise performance with background noise and, later, rotor noise.



04

Reproducible quantitative evaluation

Measure map error, sensor-fidelity effects and acoustic false-alarm rates.

Literature survey

Selected studies from the 26-work review

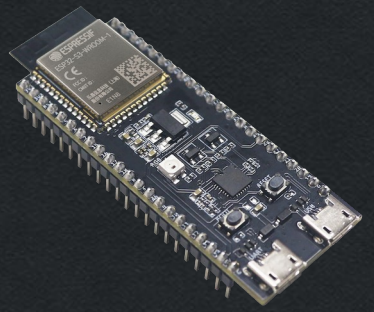
Work	Reported contribution	Implication for this project
Steux & El Hamzaoui 2010 [1]	Compact occupancy-grid SLAM	Start with a small mapping baseline
Markdahl & Vikgren 2023 [2]	tinySLAM on nano-UAVs	Evaluate mapping error quantitatively
Niculescu et al. 2024 [3]	Collaborative mapping in 1.5 MB	Embedded mapping is feasible
Tranzatto et al. 2022 [4]	Autonomous exploration and artifact search	Field reference with larger resources
Lajoie & Beltrame 2024 [5]	Sparse decentralized map sharing	Control communication cost
Sun et al. 2025 [6]	Beamforming to suppress rotor noise	Test acoustics under stated noise

Block diagram

Phase-I bench pipeline



UART ranges →



Wi-Fi relay →



Range + tilt

LiDAR and cradle

ESP32-S3

Acquisition and Wi-Fi relay

Laptop processing

Slamtec SDK and Python

Map, saved session and replay

Hardware images are references. LiDAR model confirmation pending.

Planned complete-system architecture

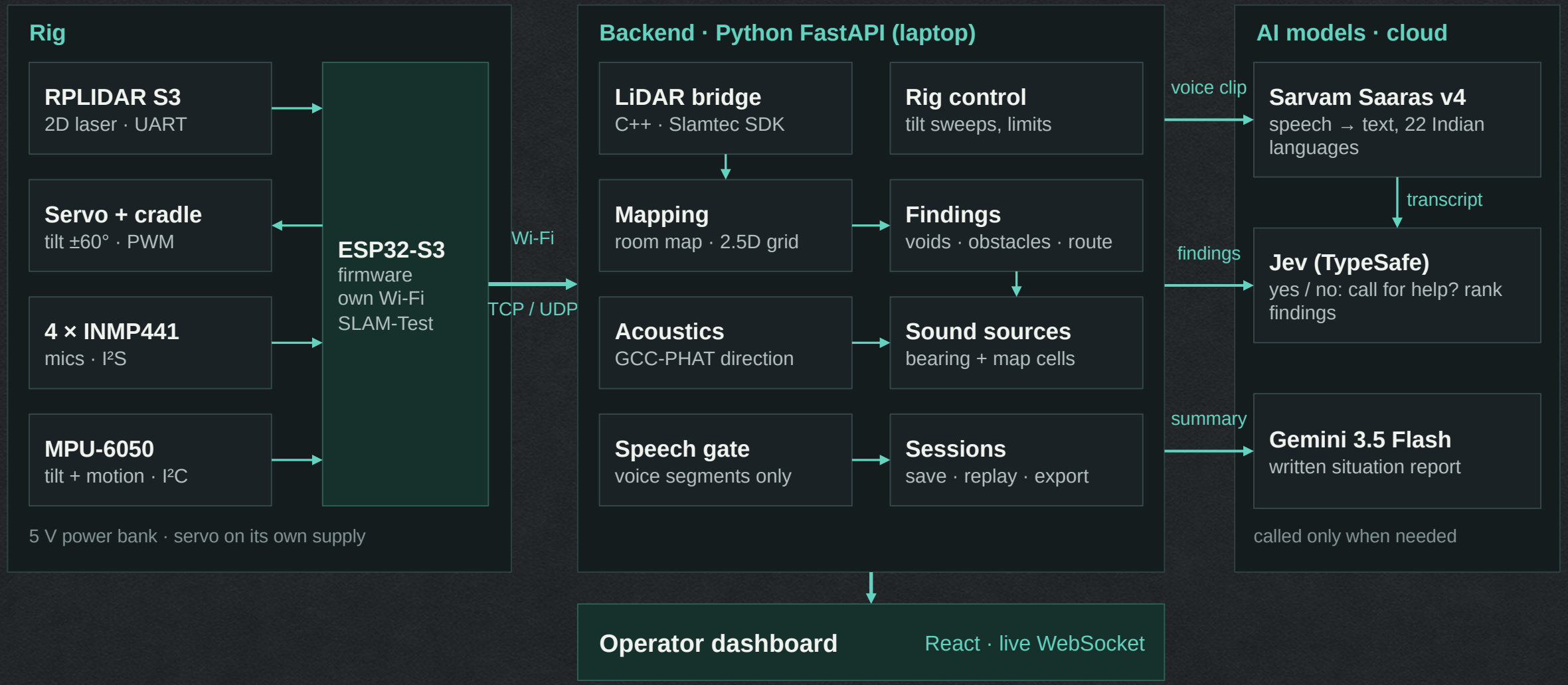
Sensor inputs → **Embedded processing** → **Offline and collaborative**

LiDAR, motion sensors and microphone array

Onboard SLAM and acoustic event / bearing

Dense 3-D reconstruction and map merging

System architecture



From the ESP32 to the backend

One Wi-Fi link carries everything; the backend splits it into simple modules

ESP32-S3 firmware

LiDAR relay
UART 1 Mbaud in, TCP port 8899 out

Tilt control
servo PWM, commands on port 8900

Microphones
4 × I²S at 16 kHz, port 8901 / 8902

Own Wi-Fi
network SLAM-Test, 192.168.4.1

Wi-Fi →

Backend server: Python 3.12 + FastAPI

LiDAR bridge
C++ on the Slamtec SDK

Rig control
tilt sweeps, safety limits

Mapping
numpy: room map, 2.5D grid

Acoustics
GCC-PHAT direction finding

Speech
Sarvam + Jev, help calls

Sessions
save, replay, PLY / CSV

↓

Dashboard: React in the browser, live over a WebSocket

Methodology







01

Review and define evidence

Choose baselines and set the evaluation protocol.

02

Calibrate the sensor and mount

Check range bias, tilt zero, direction and geometric offsets.

03

Capture controlled sweeps

Move the cradle, wait for settling, then capture each scan.

04

Reconstruct and replay

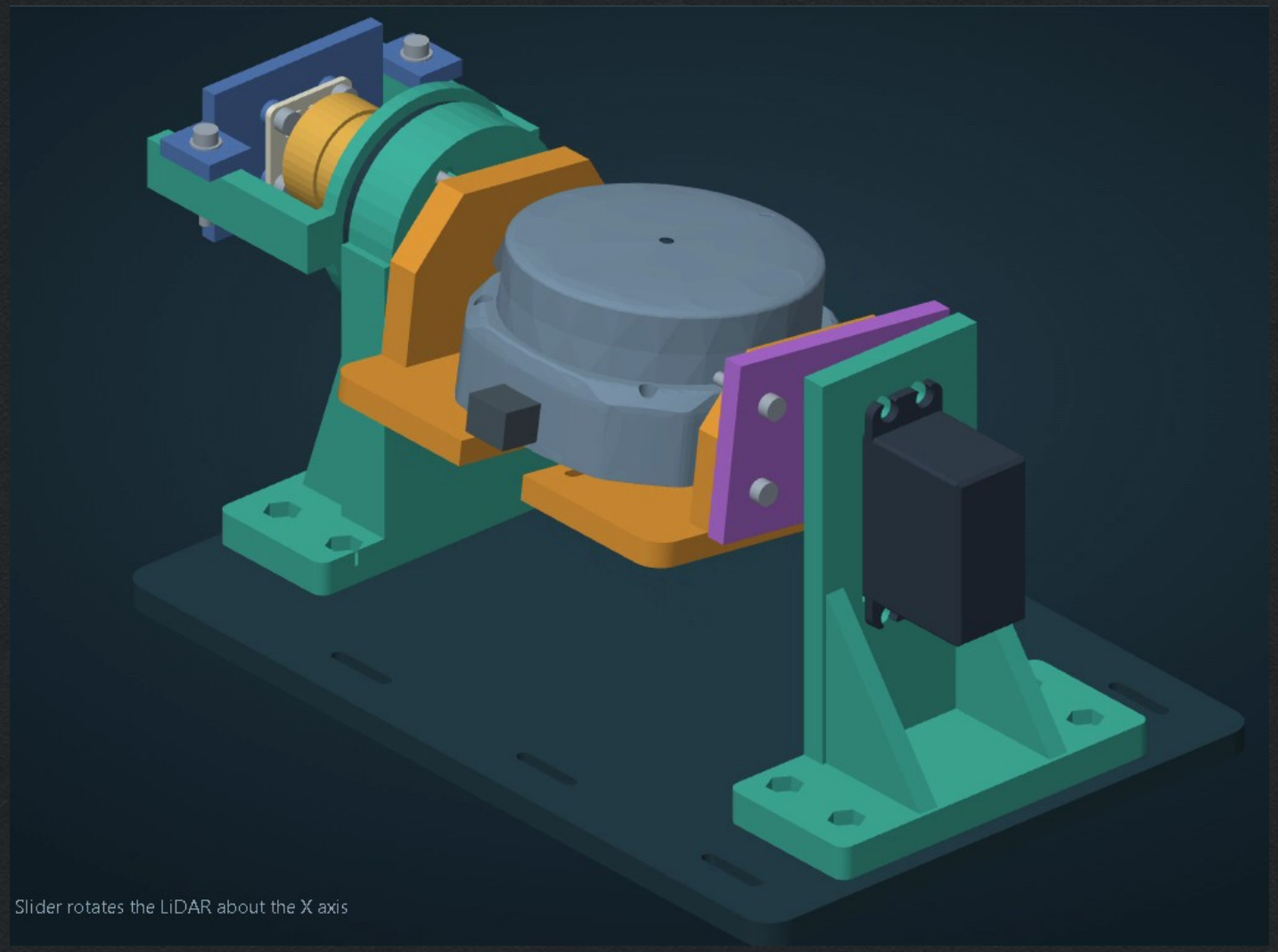
Transform points, build an elevation grid and retain raw sessions.

05

Evaluate, then extend

Compare with room measurements before acoustic, cart and flight trials.

CAD tilt mechanism



Controlled tilt adds spatial samples

The cradle rotates the LiDAR scan plane about the X axis.

Move, settle, capture

Collect each scan after the mechanism has settled.

CAD design evidence, 5.5-second animation

Components

Low-cost, off-the-shelf parts



RPLIDAR S3

2D laser · 10 Hz · 40 m

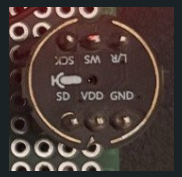
Measures distances all around in one plane



ESP32-S3 N16R8

240 MHz · 8 MB PSRAM · Wi-Fi

Brain of the rig: relays data, drives the servo



4 × INMP441

I²S MEMS microphones

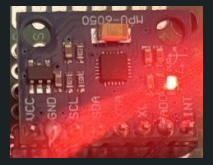
Find the direction of knocks and voices



13 kg·cm servo

metal gear · 50 Hz PWM

Tilts the LiDAR to add height



MPU-6050 IMU

accelerometer + gyro · I²C

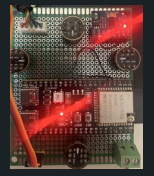
Senses tilt and motion (AS5600 encoder next)



3D-printed cradle

bearings on the X axis

Holds the LiDAR and tilts it smoothly



Carrier board

perfboard · screw terminals

Wires the ESP32, mics and connectors



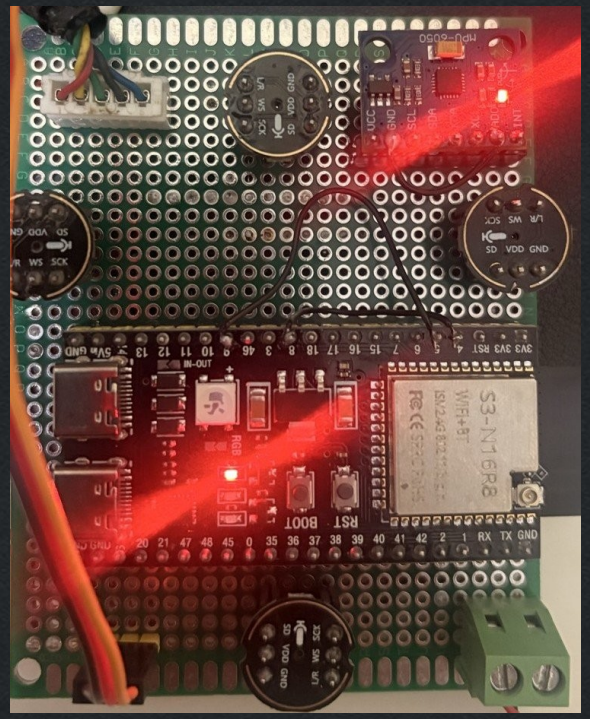
Power

5 V power bank + servo supply

Runs the rig without cables to a laptop

PCB and connections

Carrier board: every sensor lands on one ESP32-S3



Perfboard carrier: ESP32-S3 at the bottom, four microphones around it, the IMU top right, screw terminals for power.

ESP32-S3 pin	Connected to	Signal
GPIO 13 / 14	LiDAR TX / RX	UART, 1 Mbaud
GPIO 17	Servo signal	PWM, 50 Hz
GPIO 4 / 5	All mics: clock / word select	I²S
GPIO 7	Mic 1 + 2 data (left / right)	I²S
GPIO 6	Mic 3 + 4 data (left / right)	I²S
GPIO 11 / 12	IMU SDA / SCL (AS5600 next)	I²C
5 V / 3.3 V / GND	LiDAR 5 V · mics 3.3 V	power bank

Servo power comes from its own 5–6 V supply, ground shared with the ESP32.

USB-C: flashing and debug · BOOT + RESET buttons: hold BOOT, tap RESET to flash.

Phase-I results

Design and software evidence

Literature and architecture

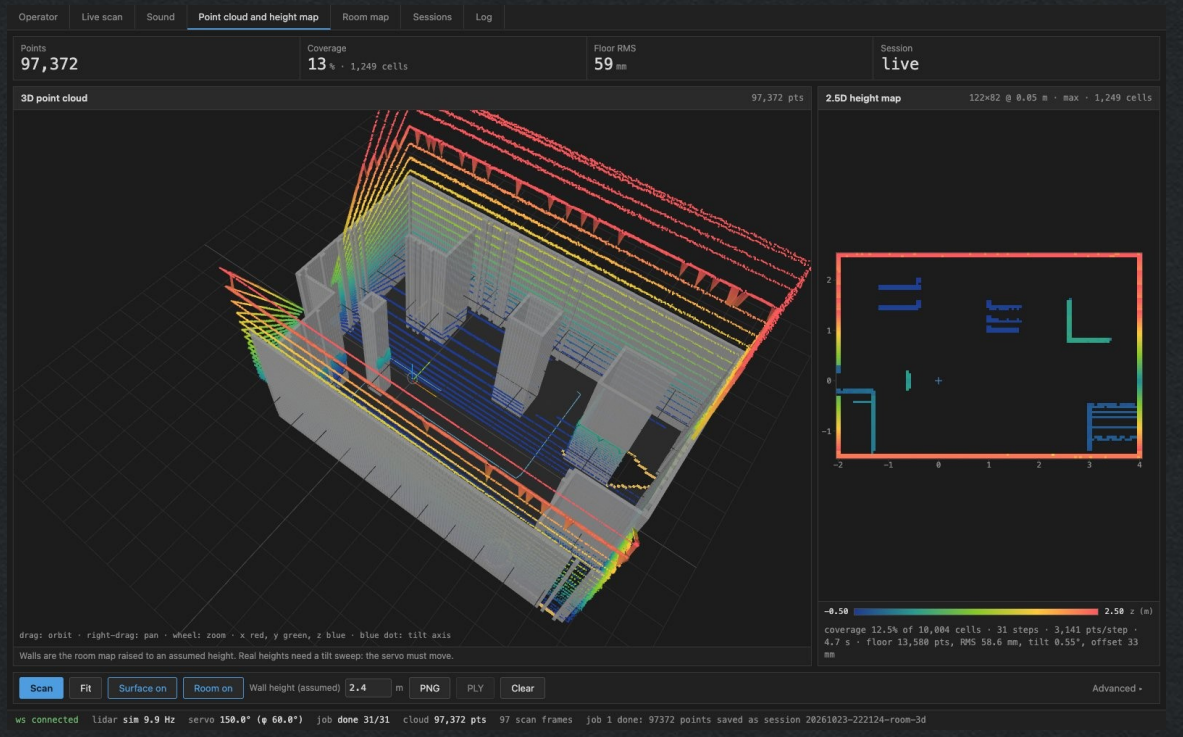
Review-based design and an evaluation plan.

Tilt mechanism and pipeline

CAD plus capture, mapping and display software.

Repeatable analysis

Raw sessions, replay and PLY / CSV export.



3D point cloud and 2.5D height map in the dashboard

Synthetic soak, 22 October

10 min Run duration

38 Completed sweeps

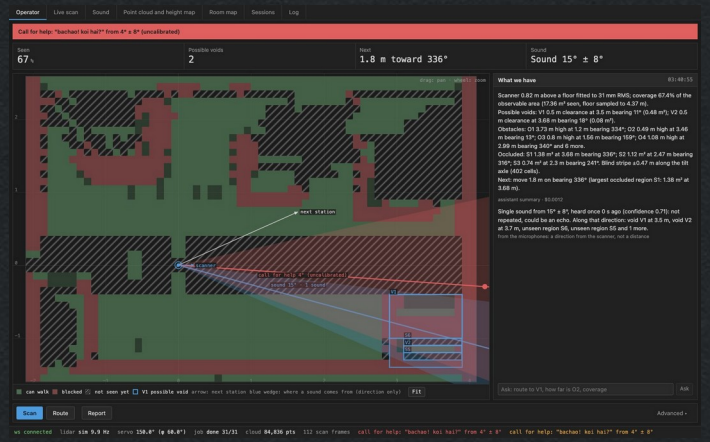
9.46 M Points processed

0 Failed starts

0 WebSocket parse errors. These synthetic results do not establish physical accuracy or acoustic performance.

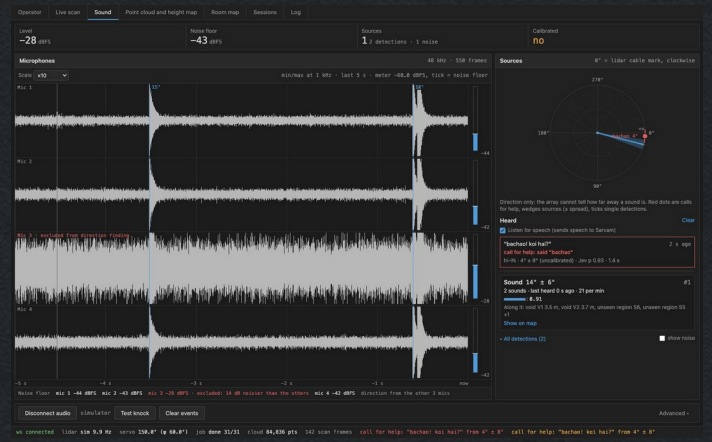
The dashboard

What each screen gives the operator



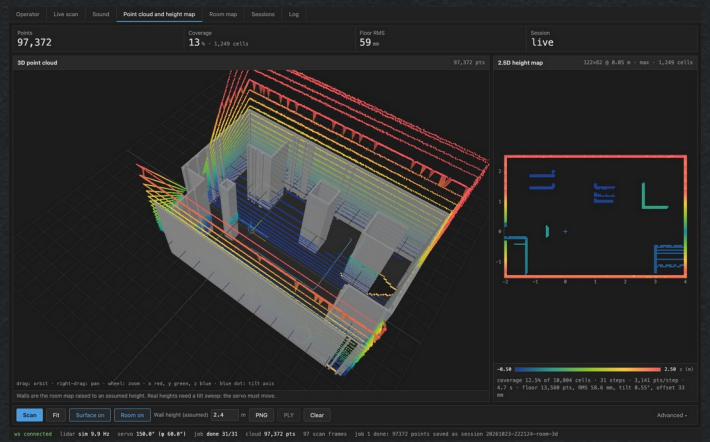
Operator

Map of walkable floor, obstacles and possible voids; next scan spot; calls for help as a red line.



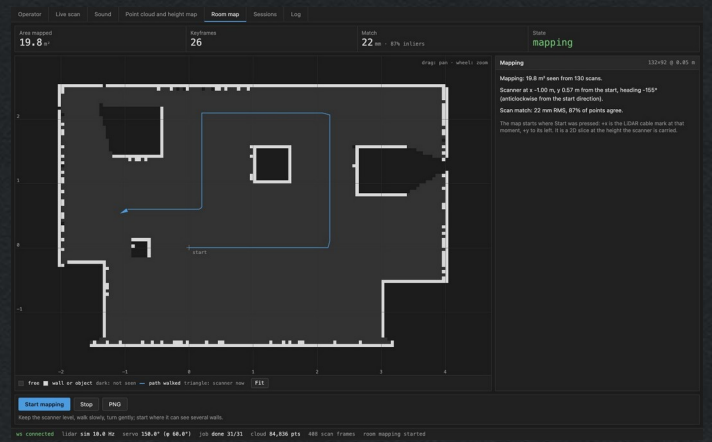
Sound

Live 4-mic waveforms, direction dial with red dots for calls for help, what was heard.



Point cloud and height map

3D view of the scanned room and the 2.5D height map below it.



Room map

Floor plan built by moving the scanner from spot to spot, with the path walked.

What makes our system different

4 mics

Hearing where people are

- Four ₹250 MEMS mics time each sound to the microsecond and turn it into a direction in about 3 ms.
- A noisy mic is dropped automatically; one clap test calibrates the array.
- “Save me”, “bachao”, “kaapadi” are flagged even when misheard, and shown as a red dot.

$0.00002

AI only where it pays

- Maps and directions are plain maths: no AI, no cost, fully explainable.
- Speech is sent only when someone talks, never for knocks or noise.
- Jev answers one yes/no question (“is this a call for help?”) instead of a chat model; Gemini only writes the final report.

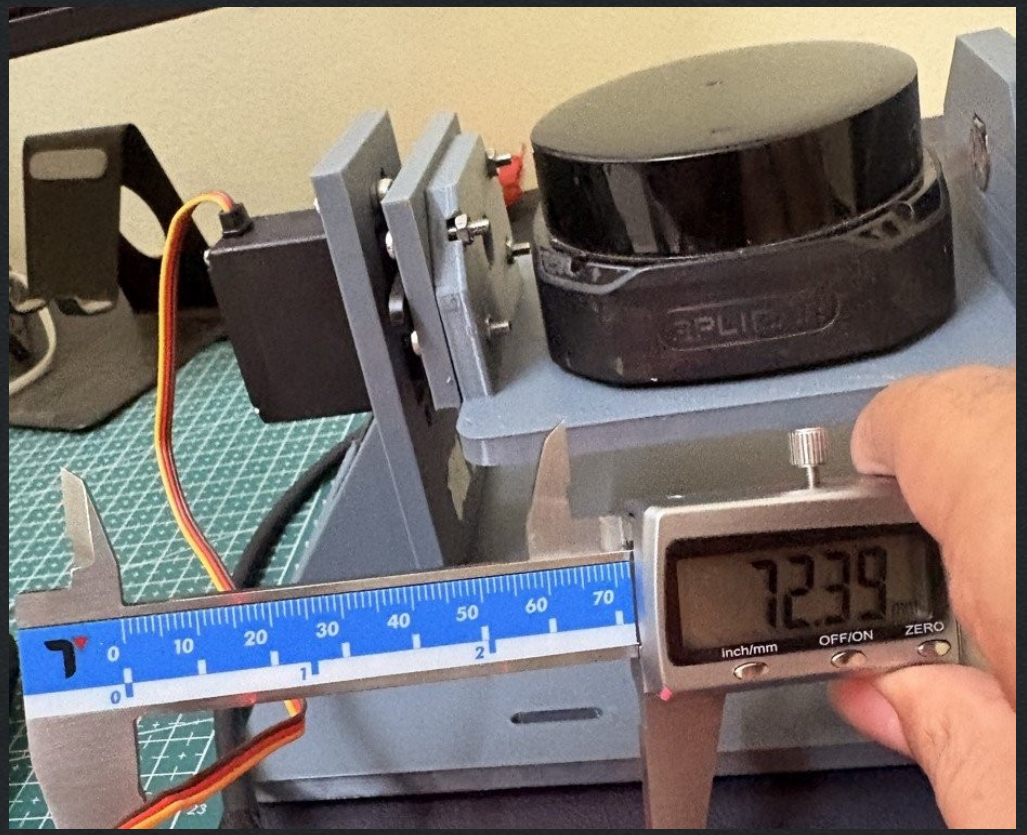
≈ ₹5k

Low-cost 3D

- A servo tilts a 2D LiDAR into a 3D scanner, instead of a 3D LiDAR costing lakhs.
- An ESP32-S3 replaces an onboard computer; the laptop does the heavy work.
- Electronics besides the LiDAR cost about ₹5,000. The same rig maps rooms by hand.

Jev: $0.00002 per call-for-help check, measured. Prices approximate, October 2026.

Project progress



Bench rig: RPLIDAR on the 3D-printed tilt cradle, with its servo.

Completed

LiDAR and 4 mics streaming over the rig's own Wi-Fi · dashboard with 7 screens · call-for-help detection with direction · room mapping from 6 spots · 125 automated tests

Measured

10 Hz, ~2,860 points per turn · range noise 0.9 mm under 1 m, 4.8 mm at 2–4 m · call for help flagged in ~2 s · map drift under 1 cm

Blockers

Servo stalls on the shared power bank: needs its own 6 V / 3 A supply · one mic picks up servo noise

Next

Separate servo supply → full 2.5D sweep at every spot · calibrate the mic array · field test in a mock collapsed room

Phase-II roadmap

From a bench rig to a field-tested 2.5D search payload

01

Nov 2026

Servo power fix

Own 6 V / 3 A supply; confirm the tilt with the IMU.

02

Dec 2026

2.5D at every station

A tilt sweep at each of the 6 spots, merged into one room model.

03

Jan 2027

Mic calibration

Clap test fixes channel order, yaw and timing; target ≤ 5° bearing error.

04

Feb 2027

Mock-rubble field test

Debris room with hidden knocks and voices; measure map error and detection rate.

05

Mar–Apr 2027

Onboard SLAM

Mapping moves onto the embedded board, toward the drone payload.

Phase-II goal: a 2.5D room model with marked voids and call-for-help directions, checked against the real room.

References

[1] Steux & El Hamzaoui (2010)

tinySLAM: A SLAM algorithm in less than 200 lines C-language program

ICARCV, pp. 1975-1979

[2] Markdahl & Vikgren (2023)

tinySLAM-based exploration with a swarm of nano-UAVs

SWARM 2023 / arXiv:2309.02834

[3] Niculescu et al. (2024)

Ultra-Lightweight Collaborative Mapping for Robot Swarms

arXiv:2407.03136

[4] Tranzatto et al. (2022)

CERBERUS in the DARPA Subterranean Challenge

Science Robotics 7, eabp9742

[5] Lajoie & Beltrame (2024)

Swarm-SLAM: Sparse Decentralized Collaborative SLAM Framework

IEEE RA-L 9(1), 475-482

[6] Sun et al. (2025)

Acoustic event detection for drone search and rescue using rotor-noise beamforming

Digital Signal Processing 157, 104881

Publication details

Publication plan

Venue

Review paper drafted; venue to be finalised.

Status

In preparation, after Phase-II results.

Date and link

To be added on submission.



Draft review paper

Onboard SLAM, 3-D collaborative mapping and acoustic survivor detection: a review.